

BALGUNTAY, ChinaWMO: **514670**Lat: **42.669N**Lon: **86.329E**Elev: **1733**StdP: **82.16**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-16.3	-28.8	0.3	-11.1	-26.7	0.4	-10.9	6.2	-6.4	5.6	-6.8	1.0	90	0.485

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	28.7	14.1	27.4	13.7	26.1	13.3	16.0	24.6	15.1	24.0	14.4	23.3	2.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	11.7	18.2	12.1	10.9	17.2	11.1	10.1	17.6	51.3	24.6	48.5	23.9	46.3	23.5	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.2	4.5	-19.6	31.3	2.3	1.3	-21.3	32.2	-22.6	33.0	-23.9	33.7	-25.6	34.6		
(5)			-20.3	17.0	2.1	1.3	-21.9	17.9	-23.1	18.7	-24.3	19.4	-25.8	20.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.6	-7.7	-3.4	3.4	10.0	14.6	18.1	19.6	18.8	14.6	8.1	1.0	-6.1
	DBStd	10.15	3.82	3.72	4.08	4.22	3.53	2.99	2.53	2.62	3.04	3.57	3.87	3.70
	HDD10.0	2040	547	376	208	51	7	0	0	0	4	82	268	498
	HDD18.3	4042	806	609	464	251	122	40	15	26	117	319	518	756
	CDD10.0	1179	0	0	2	51	150	242	298	274	141	21	0	0
	CDD18.3	138	0	0	0	1	6	32	55	41	4	0	0	0
	CDH23.3	1188	0	0	0	9	63	251	466	357	42	1	0	0
	CDH26.7	191	0	0	0	1	3	33	96	57	1	0	0	0
Wind	WSAvg	1.9	1.8	2.1	2.3	2.2	2.0	1.7	1.5	1.6	1.6	1.7	1.9	1.7
Precipitation	PrecAvg	193	1	0	2	4	25	40	45	39	28	4	0	0
	PrecMax	289	5	4	28	20	76	89	131	141	100	23	2	3
	PrecMin	111	0	0	0	0	1	0	7	1	1	0	0	0
	PrecStd	50	1	1	5	5	19	27	33	30	21	6	1	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	9.1	17.6	24.5	27.0	29.4	31.2	29.9	26.4	21.3	13.7	4.7
		MCWB	-1.9	0.7	5.9	9.9	11.5	13.5	15.4	14.5	13.1	9.3	4.6	-1.4
	2%	DB	1.9	6.5	14.9	21.8	25.2	27.6	29.1	28.2	24.5	18.7	11.0	2.4
		MCWB	-3.5	-0.8	4.5	8.6	10.8	13.0	14.7	14.2	12.0	8.1	2.8	-2.9
	5%	DB	-0.1	4.6	12.9	19.7	23.5	26.2	27.5	26.9	22.9	16.7	8.8	0.7
		MCWB	-4.8	-1.8	3.4	7.6	10.3	12.7	14.2	13.9	11.2	7.0	1.6	-4.1
	10%	DB	-2.0	2.8	10.8	17.8	21.8	24.7	26.0	25.4	21.3	14.8	6.9	-0.9
		MCWB	-6.1	-2.9	2.3	6.8	9.7	12.3	14.0	13.4	10.6	5.9	0.5	-5.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.7	1.0	6.3	10.5	12.4	15.1	17.7	17.2	14.1	9.8	4.8	-1.3
		MCDB	3.9	8.8	16.7	22.9	24.3	24.6	25.0	24.4	24.0	20.0	13.1	3.8
	2%	WB	-3.4	-0.6	4.8	9.2	11.6	14.2	16.4	15.7	12.8	8.5	3.0	-2.7
		MCDB	1.7	6.0	14.1	20.3	23.3	23.8	24.8	24.2	22.0	17.5	10.3	2.1
	5%	WB	-4.7	-1.7	3.6	8.1	10.8	13.6	15.5	14.8	11.9	7.5	1.8	-3.9
		MCDB	-0.2	4.2	12.2	18.4	22.0	23.0	24.3	23.7	21.0	15.7	8.2	0.4
	10%	WB	-6.0	-2.8	2.5	7.2	10.1	13.1	14.8	14.2	11.0	6.4	0.7	-5.0
		MCDB	-2.1	2.5	10.4	16.7	20.7	22.2	23.5	23.3	19.8	13.9	6.6	-1.1
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.9	12.2	11.8	11.3	11.0	11.4	11.9	12.1	11.9	10.0	9.0
		MCDBR	11.9	13.1	14.4	14.1	13.4	13.1	13.8	14.1	14.1	13.8	11.7	9.8
		MCWBR	8.1	8.2	7.7	6.5	5.2	4.5	4.7	5.1	5.6	6.4	6.6	6.5
	5% WB	MCDBR	11.6	12.7	13.6	13.0	12.1	11.4	11.7	12.0	12.7	12.7	11.2	9.7
		MCWBR	7.9	8.0	7.2	6.0	4.8	4.2	4.6	5.0	5.3	6.0	6.4	6.5
Clear-Sky Solar Irradiance	taub	0.284	0.338	0.416	0.482	0.492	0.502	0.468	0.461	0.435	0.379	0.318	0.280	
	taud	2.264	2.108	1.915	1.799	1.834	1.871	2.003	2.003	2.006	2.075	2.196	2.279	
	Ebn at Noon	849	842	809	785	791	783	806	796	783	780	787	817	
	Edh at Noon	88	123	169	206	204	198	171	166	153	125	94	79	
All-Sky Solar Radiation	RadAvg	2.30	3.33	4.63	5.74	6.27	6.22	6.20	5.79	5.02	3.76	2.54	1.95	
	RadStd	0.13	0.18	0.20	0.33	0.32	0.45	0.32	0.34	0.18	0.20	0.15	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-1.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92	+52

Nomenclature: See separate page